



10655 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Tea Temim (PI)	Princeton University
Dr. Patrick Slane (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Dan Milisavljevic (CoI)	Purdue University
Dr. Parviz Ghavamian (CoI)	Towson University
Dr. Nathan Smith (CoI)	University of Arizona
Dr. Richard G. Arendt (CoI)	University of Maryland Baltimore County
Dr. Hans-Thomas Janka (CoI) (ESA Member)	Max Planck Institute for Astrophysics
Prof. Ilse De Looze (CoI) (ESA Member)	Universiteit Gent
Dr. John Charles Raymond (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Robert A. Fesen (CoI)	Dartmouth College
Joseph Gelfand (CoI)	New York University
Dr. Ravi Sankrit (CoI)	Space Telescope Science Institute
Dr. Brian Williams (CoI)	NASA Goddard Space Flight Center
Dr. Ori Dosovitz Fox (CoI)	Space Telescope Science Institute
Dr. Paul P. Plucinsky (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Armin Rest (CoI)	Space Telescope Science Institute
Dr. Daniel J. Patnaude (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Salvatore Orlando (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Palermo
Dr. P. Frank Winkler (CoI)	Middlebury College
J. Martin Laming (CoI)	Naval Research Laboratory

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
G292.0+1.8				
	1	MIRI Imaging Mosaic	MIRI Imaging	(12) SNR-G292.0+01.8
	2	MRS pointing 1	MIRI Medium Resolution Spectroscopy	(20) SNR-G292.0+01.8-IFU-1
	3	MRS pointing 2	MIRI Medium Resolution Spectroscopy	(21) SNR-G292.0+01.8-IFU-2
	4	MRS pointing 3	MIRI Medium Resolution Spectroscopy	(23) SNR-G292.0+01.8-IFU-3
	5	MRS bkg	MIRI Medium Resolution Spectroscopy	(13) SNR-G292.0+01.8-BKG

ABSTRACT

Core-collapse supernovae (SNe) are the primary source of heavy elements in the Universe and play a crucial role in shaping the energy balance, structure, and chemical composition of galaxies. However, key aspects of these explosions -- such as the nature of the progenitor systems and the effects of late-stage stellar mass loss on the evolution of the expelled material -- remain poorly understood. Stripped-envelope SNe, which result from stars that lose their outer hydrogen envelopes through strong stellar winds or binary interactions, account for 30% of core-collapse SNe and are among the least understood explosion types.

We propose JWST observations of the well-known supernova remnant G292.0+1.8 that resulted from a stripped-envelope SN explosion. Its evolutionary stage offers a unique advantage: Most of its circumstellar medium (CSM) and supernova-ejected material have been shocked, providing a comprehensive view of the progenitor's mass-loss history and the explosion asymmetries and dynamics. Leveraging JWST's unparalleled spatial resolution and sensitivity, we will map the CSM and ejecta structures in unprecedented detail. With the help of 3D magnetohydrodynamic simulations, the JWST observations will allow us to address key questions about progenitor's mass loss, explosion properties, dust survival, and the pulsar's kick mechanism.

OBSERVING DESCRIPTION

The SNR is mosaicked with the MIRI imager using the 4-point-extended-source dither pattern. A 2x1 mini mosaic is linked to be executed within 53 days within the main mosaic. A dedicated background observation is linked with the mini mosaic as a non-interruptible sequence. MIRI MRS observations will be acquired in three locations using the extended source dither pattern. The precise coordinates for the MRS pointings will be determined once the imaging observations have been acquired. We have, therefore, added an "ON HOLD" constraint to the MRS observations. A dedicated background for MRS observations is included and grouped as a non-interruptible sequence.

Proposal 10655 - Observation 1 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Observation	Proposal 10655, Observation 1: MIRI Imaging Mosaic	Tue Jun 02 15:00:43 GMT 2026
	Diagnostic Status: Warning	
	Observing Template: MIRI Imaging	

[illegible]

Proposal 10655 - Observation 1 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

	(Visit 1:46) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(12)	SNR-G292.0+01.8	RA: 11 24 31.0935 (171.1295563d) Dec: -59 15 52.78 (-59.26466d) Equinox: J2000			Epoch of Position: 2015.5				
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	Category=ISM Description=[Supernova remnants] Extended=YES									
Template	Subarray									
	FULL									
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order
	6	8	10.0		8.0		0.0		0.0	HILBERT_CURVE
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
	1	4-Point-Sets				6	1	EXTENDED SOURCE	POSITIVE	DEFAULT
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time
	1	F770W	FASTR1	15	1	1	Dither 1	4	4	166.502
	2	F1280W	FASTR1	35	1	1	Dither 1	4	4	388.506
	3	F1800W	FASTR1	20	1	1	Dither 1	4	4	222.003
	4	F2100W	FASTR1	20	2	1	Dither 1	4	8	455.107
	5	F2550W	FASTR1	10	4	1	Dither 1	4	16	477.307
Special Requirements	Sequence Visits within 53.0 Days Visits Same PA									
	2 After 1 by 60 Days to 150 Days									
	3 After 1 by 60 Days to 150 Days									
	4 After 1 by 60 Days to 150 Days									
	5 After 1 by 60 Days to 150 Days									

Proposal 10655 - Observation 2 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Observation	Proposal 10655, Observation 2: MRS pointing 1												Tue Jun 02 15:00:43 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MRS bkg (Obs 5)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(20)	SNR-G292.0+01.8-IFU-1	RA: 11 24 47.8905 (171.1995438d) Dec: -59 15 40.62 (-59.26128d) Equinox: J2000				Epoch of Position: 2015.5							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=ISM													
	Description=[Supernova remnants]													
Acquisition	Extended=YES													
	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	1	LONG(C)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	1	LONG(C)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	2		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	3		IMAGER	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	3	SHORT(A)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	3	SHORT(A)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		

Proposal 10655 - Observation 2 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Special Requirements	2 After 1 by 60 Days to 150 Days Sequence Observations 5, 2, 3, 4 (reordered), Non-interruptible
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Proposal 10655 - Observation 3 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Observation	Proposal 10655, Observation 3: MRS pointing 2												Tue Jun 02 15:00:43 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MRS bkg (Obs 5)]												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(21)	SNR-G292.0+01.8-IFU-2	RA: 11 24 27.2992 (171.1137467d)				Epoch of Position: 2015.5						
			Dec: -59 15 50.52 (-59.26403d)										
			Equinox: J2000										
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES											
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	1	LONG(C)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224	
	1	LONG(C)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224	
	2		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224	
	3		IMAGER	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	3	SHORT(A)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224	
	3	SHORT(A)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224	

Proposal 10655 - Observation 3 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Special Requirements	3 After 1 by 60 Days to 150 Days Sequence Observations 5, 2, 3, 4 (reordered), Non-interruptible
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Proposal 10655 - Observation 4 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Observation	Proposal 10655, Observation 4: MRS pointing 3												Tue Jun 02 15:00:43 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MRS bkg (Obs 5)]													
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(23)	SNR-G292.0+01.8-IFU-3			RA: 11 24 18.8452 (171.0785217d)			Epoch of Position: 2015.5						
					Dec: -59 13 51.65 (-59.23101d)									
					Equinox: J2000									
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES												
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type						Optimized For			Direction			
	1	4-Point						EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	1	LONG(C)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	1	LONG(C)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	2		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	3		IMAGER	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	3	SHORT(A)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	3	SHORT(A)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		

Proposal 10655 - Observation 4 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Special Requirements	4 After 1 by 60 Days to 150 Days Sequence Observations 5, 2, 3, 4 (reordered), Non-interruptible
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Proposal 10655 - Observation 5 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Observation	Proposal 10655, Observation 5: MRS bkg												Tue Jun 02 15:00:43 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MRS pointing 1 (Obs 2), MRS pointing 2 (Obs 3), MRS pointing 3 (Obs 4)]													
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(13)	SNR-G292.0+01.8-BKG	RA: 11 25 18.6106 (171.3275442d)				Epoch of Position: 2015.5							
			Dec: -59 10 52.90 (-59.18136d)											
			Equinox: J2000											
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES												
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	1	LONG(C)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	1	LONG(C)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	2		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	3		IMAGER	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504		
	3	SHORT(A)	MRSLONG		FASTR1	50	3	1	Dither 1	4	12	1687.224		
	3	SHORT(A)	MRSSHORT		FASTR1	50	3	1	Dither 1	4	12	1687.224		

Proposal 10655 - Observation 5 - A JWST Study of the Stripped-Envelope Supernova Remnant G292.0+1.8

Special Requirements	5 After 1 by 60 Days to 150 Days Sequence Observations 5, 2, 3, 4 (reordered), Non-interruptible
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